

Udayan Atreya

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EDUCATION

Master of Science, Computer Science

May 2026

San Jose State University, San Jose, US

Coursework: Applied Deep Learning, Retrieval Augmented Generation (RAG), Biometric Security.

Bachelor of Technology, Computer Science & Engineering

May 2020

Manipal University, Jaipur, India

Coursework: Data Structures & Algorithms, Database Management Systems, Object-Oriented Programming.

TECHNICAL SKILLS

Languages: Python, Java, C#, JavaScript, HTML, CSS, SQL

Web & Frameworks: FastAPI, .NET Core, Spring MVC, REST APIs, JWT, LangChain, HuggingFace, Git

LLMs & AI Tools: Claude, GPT, Gemini, Ollama, Llama, TensorFlow, PyTorch, RAGAS

Cloud & Database: AWS (S3, Lambda, API Gateway), CI/CD, MongoDB, DynamoDB, Pinecone, ChromaDB

PROJECT EXPERIENCE

WeatherAPI (C#, .NET Core, REST API, JWT, MongoDB)

[\[GitHub Link\]](#)

- Designed and built a production-style RESTful Web API in .NET Core following a clean layered architecture (Controllers, Services, Models), exposing well-documented, maintainable endpoints ready for frontend and mobile client integration.
- Implemented dual-mode weather retrieval—city-specific queries and automatic IP geolocation-based lookup—integrating third-party backend APIs to deliver seamless user experiences without client-side configuration.
- Built JWT-based user authentication (registration, login, token issuance and validation) backed by MongoDB for persistent data storage, enabling stateless, scalable, and secure API access for consuming web frontends.

BobTheAgent (Python, CrewAI, HTML, Computer Vision, Blockchain)

[\[GitHub Link\]](#)

- Built a live HTML dashboard surfacing real-time construction project status to clients, builders, and subcontractors—a responsive, user-facing web interface that eliminated information asymmetry across all stakeholders.
- Orchestrated a network of six specialized AI agents (Client, Bidding, Scheduling, Vision, Payment, Master) using CrewAI, automating the full project lifecycle from bid evaluation through milestone-based blockchain payment release.
- Integrated a Computer Vision AI model for passive site-progress monitoring and wired its verified output into the scheduling and payment pipeline, triggering trustless automated payments via Masumi Cardano blockchain.

WORK EXPERIENCE

Software Engineer, Accenture, Bengaluru, India

Dec 2021 – Jul 2024

Associate Software Engineer, Accenture, Bengaluru, India

Nov 2020 – Nov 2021

- Architected a user-facing notification system exposing REST APIs (AWS API Gateway, Lambda, DynamoDB) to deliver real-time remediation alerts to web clients, **increasing DAU by 20%** post-launch.
- Built and integrated backend APIs and microservices for a ChatOps web platform, handling data exchange between frontend and automation workflows and **reducing overhead by 40%** (~5m to ~1m).
- Optimized DynamoDB access patterns and CRUD API microservices powering web-facing dashboards, **reducing retrieval latency by 37%** and ensuring resilient, high-performance data delivery to frontend clients.
- Delivered a code vulnerability remediation solution via serverless microservices (AWS Lambda, S3, EventBridge) and Azure DevOps CI/CD, with clean well-documented code safeguarding **400+ applications** across organizations.
- Designed a single-touch server patching solution using serverless architecture (AWS, Azure DevOps) for automated alert resolution, **reducing MTTR by 80%**.

Software Intern, Motherson Technologies, India

Dec 2019 – Apr 2020

- Designed and built a Java Spring MVC web dashboard as the primary stakeholder-facing UI for financial analytics, processing and visualizing **200,000+ tax receipts** with interactive, filterable data views.
- Developed interactive JavaScript data visualization modules integrated with a PostgreSQL backend, enabling real-time filtering and drill-down reporting that **accelerated audit resolution by 14%**.
- Implemented clean MVC separation between the web presentation layer, business logic, and data access, replacing manual review workflows with structured, responsive, and maintainable dashboard views.